Copilot in Excel: data quality and SPC

140 minutes

IES | Original synthetic CW-401 engineering case. Values and limits are fictional teaching assumptions, not plant operating instructions. Generated references are independently calculated examples, not recorded Copilot executions.

Preparation

Save a working copy of Starter.xlsx to your approved OneDrive or SharePoint location. Open it in Excel with your work/school account and use the Copilot side pane. Select the named table before prompting. If Copilot cannot edit, request the formula and insert it manually; record the fallback. Confirm tenant entitlement and supported workbook format before class. Keep the local raw files unchanged.

1. Prepare the workbook

Save a working copy of Starter.xlsx to your approved OneDrive or SharePoint location. Open it in Excel with your work/school account and use the Copilot side pane. Select the named table before prompting. If Copilot cannot edit, request the formula and insert it manually; record the fallback. Confirm tenant entitlement and supported workbook format before class. Keep the local raw files unchanged. Open ReadMe, Parameters, RawTests and Prompts. The workbook starts with source tables and blank output sheets.

2. Audit without deleting evidence

Select RawTests and submit prompt SPC1 from Prompts. Inspect TEST-068 (missing volume), TEST-083 (110-second invalid test) and duplicate TEST-088. Keep one copy of the duplicate; exclude the two invalid tests from the comparable analysis. Retain121 raw rows,120 unique tests and118 valid tests. Write decisions into QualityAudit.

3. Build the analysis table

Ask Copilot to filter and sort the valid unique records into Analysis. ValidTests is the prepared comparison source if cleaning stalls. Keep test_id, timestamp, phase, shift, pump, collected_L and differential pressure. Add delivery_L_h using collected_L*3600/Parameters duration. Spot-check one baseline and one monitoring row with a calculator.

4. Calculate a fixed baseline

Submit SPC2. Only60 baseline records estimate the centre and moving ranges. First baseline MR is blank; next59 are absolute successive differences. Mean62.00250 L/h; MRbar0.73881; sigma0.65498; LCL60.03757; UCL63.96743. Compare formulas with the reference after attempting the task.

5. Chart process behaviour

Submit SPC3. Use time order and phase labels; add constant mean, UCL and LCL series. Keep baseline limits for monitoring and pilot. There are28 valid monitoring and30 pilot tests. Seventeen monitoring points TEST-073–090 except083 are outside the limits. The missing record and invalid duration remain in the quality ledger.

6. Discuss capability and evidence

Submit SPC4. Conditional baseline Cp1.57766 and Cpk1.57639 use supplied specs. Check baseline behaviour and measurement assumptions before interpreting them. Do not transfer that conclusion to the shifted monitoring period. Note the differential-pressure association and the two changes before pilot; neither establishes a sole cause.

7. Retain the result

Save your workbook, prompts, chart, quality decisions, checks and a100-word interpretation. Record what Copilot actually did and what you calculated manually. Use the lab evaluation checklist and compare with the worked reference only after your attempt.

Copilot prompts

SPC1 Audit the source

Using RawTests, audit test_id uniqueness, missing collected_L, duration_s and quality_flag. Do not modify RawTests. Create a QualityAudit table with record ID, issue, inclusion decision and reason. Use one unique record per test, require VALID and 120 seconds, and never discard an excursion solely because it is extreme. Report raw, unique and valid counts.

SPC2 Calculate the baseline

Use ValidTests in time order. In Analysis create delivery_L_h = collected_L*3600/120. Use only the 60 baseline records to calculate mean and successive moving ranges; do not bridge across phases. Estimate sigma = MRbar/1.128 and individuals limits mean +/-3*sigma. Put constants and specifications in Parameters and show formulas, units and referenced rows.

SPC3 Plot and interpret

Create an Excel line chart of delivery_L_h by timestamp for baseline, monitoring and pilot, with the fixed baseline mean/UCL/LCL. Flag points outside the limits without refitting them. Report phase counts and affected IDs. Explain the monitoring shift as a signal requiring investigation, not a proven restriction diagnosis. Keep invalid and missing tests visible in QualityAudit.

SPC4 Check capability

Using fictional test specifications 58.9–65.1 L/h, calculate baseline Cp and Cpk from the stated within-process estimator. State assumptions: stable baseline, adequate measurement, suitable distribution and independence. Do not use a good Cp to claim the later monitoring period is stable. Review potential shift/measurement effects and compare with Worked-Reference.xlsx.